DATHEON PYTHON INTERNSHIP

4 Python Projects

Built from scratch with the standard library

1	**Hangman Game**	ASCII art, 6 lives
2	**File Organizer**	Auto-sorts messy folders
3	**Support Chatbot**	Keywords + regex
4	**Portfolio Tracker**	P/L report + CSV log

#Python #Datheon #Internship

TASK 1

Hangman Game

Terminal game with live ASCII gallows

+ Random word from a 10,000-word English list
+ 6 wrong attempts, ASCII art per mistake
+ Tracks guesses, no penalty for repeats

```
Good guess!

     +---+
     |   |
     O   |
     |   |
         |
         |
    =========

Word   : P L A N E _
Guessed: A E L N P X Z
Lives  : 4
Guess a letter: t
Good guess!

     +---+
     |   |
     O   |
     |   |
         |
         |
    ========

You won! The word was 'PLANET'.
```

#Python #Datheon #Internship

TASK 2

File Organizer

Cleans cluttered folders automatically

+ Creates Documents/Images/Videos/Archives/Code
+ Handles name collisions and permission errors
+ Dry-run mode + execution summary

```
==========================================
 ORGANIZER SUMMARY
==========================================
 Folder        : ~/Downloads
 Files moved   : 9
   - Archives  : 1
   - Code      : 2
   - Documents : 2
   - Images    : 2
   - Others    : 1
   - Videos    : 1
 Skipped       : 0
 Errors        : 0
==========================================

Result:
Archives/ -> backup.zip
Code/ -> app.py, index.html
Documents/ -> notes.txt, report (1).pdf, report.pdf
Images/ -> photo1.jpg, photo2.png
Others/ -> data.xyz
Videos/ -> movie.mp4
```

#Python #Datheon #Internship

TASK 3

Support Chatbot

Rule-based assistant using keywords and regex

+ Hours, prices, shipping, returns, order tracking
+ Fallback for unknown queries
+ Clean exit with bye / exit / quit

```
TechMart Assistant (type 'bye', 'exit' or 'quit' to leave)
------------------------------------------------------
You: hello
Bot: Hello! Welcome to TechMart. How can I help you today?
You: what are your business hours?
Bot: We're open 9:00 AM - 8:00 PM (Mon-Fri) and 10:00 AM - 6:00 PM (Sat-Sun).
You: how much does shipping cost
Bot: Standard delivery takes 3-5 business days. Free shipping on orders above Rs. 999.
You: I want a refund
Bot: You can return any product within 7 days of delivery. Refunds take 5-7 working
      days.
You: where is ORD12345
Bot: Order ORD12345 is being processed and will be dispatched within 24 hours.
You: blah blah
Bot: Sorry, I didn't understand that. You can ask about business hours, prices,
      shipping, returns, warranty, payment or order tracking.
You: bye
Bot: Thanks for visiting! Goodbye.
```

#Python #Datheon #Internship

TASK 4

Portfolio Tracker

Investment P/L report logged to CSV

+ Ticker, buy price and quantity input
+ Per-holding and total gain/loss %
+ Auto-saves portfolio_summary.csv

```
Enter your holdings (leave ticker empty to finish).
Ticker symbol: AAPL
  Buy price: 150
  Quantity : 10
Ticker symbol: MSFT
  Buy price: 400
  Quantity : 5
Ticker symbol: NVDA
  Buy price: 100
  Quantity : 20
Ticker symbol:
================================================================================
Ticker       Qty       Buy       Now    Invested       Value     P/L %
--------------------------------------------------------------------------------
AAPL          10    150.00    225.50     1500.00     2255.00   +50.33%
MSFT           5    400.00    415.20     2000.00     2076.00    +3.80%
NVDA          20    100.00    135.60     2000.00     2712.00   +35.60%
--------------------------------------------------------------------------------
Total invested: 5,500.00 | Portfolio value: 7,043.00 | Overall: +1,543.00 (+28.05%)
================================================================================
Transaction summary saved to portfolio_summary.csv
```

WHAT I LEARNED

Key takeaways

+ File handling with os, shutil and pathlib
+ Regex and rule-based conversation flow
+ Error handling: try / except
+ Working with CSV and real datasets
+ Writing clean, reusable functions

Thank you @Datheon for the opportunity!

GitHub: github.com/<your-username>/datheon-python-internship

#Python #Datheon #Internship